Narrative Shift Detection in News Media using Temporal Contrastive Learning

What is a Narrative?

Framing & Interpretation

A narrative in news media refers to how events, entities, and issues are **framed, structured, and interpreted over time**. It is not just about reporting facts, but about how those facts are organized with tone, emphasis, and context to convey a particular meaning. Narratives evolve as real-world situations change, reflecting shifts in political, social, and informational contexts.

Beyond Raw Facts

Narratives go beyond isolated facts by **connecting multiple events into a coherent temporal story**. Even when discussing the same topic, different articles may present different interpretations due to changes in context, perspective, or emphasis. This makes narrative a **dynamic semantic phenomenon**, where meaning changes gradually or abruptly rather than remaining static.

Public Perception

Narratives directly influence how audiences understand events by shaping **public perception, opinion, and discourse**. As new developments occur, narratives shift—for example, moving from “threat” to “crisis” to “recovery” in evolving news coverage. Detecting these shifts is essential for analyzing media bias, tracking discourse evolution, and understanding how information impacts society.

What is Narrative Shift?

A **narrative shift** occurs when the dominant framing or interpretation of a topic changes across time — driven by real-world developments, editorial decisions, or evolving public discourse. Narrative shifts are not just changes in facts, but changes in the **interpretation of those facts**, reflecting evolving real-world situations, public discourse, and media focus.

→ Shifts may be **gradual** — evolving incrementally as new information accumulates — or **abrupt**, triggered by sudden events.

→ The same topic can carry **entirely different semantic framing** across distinct time periods without any change in the underlying subject matter.

→ Detecting such shifts provides a systematic lens for understanding **evolving media discourse** at scale.

Real-World Example: COVID-19 Narrative Evolution

1

Early Coverage

Framing centered on **origin and localized spread** — treating the outbreak as a regional health concern.

2

Mid-Stage Coverage

Narrative shifted to a **global crisis** frame, emphasizing cross-border spread and overwhelmed healthcare systems.

3

Late-Stage Coverage

Focus transitioned to **vaccination, recovery, and economic reopening** — marking a clear interpretive transformation.

Why Narrative Shift Detection Matters

Media Bias Tracking

Narrative shift detection helps uncover how **media framing and bias evolve over time**. By analyzing changes in tone, emphasis, and interpretation across news sources, it becomes possible to identify systematic patterns in coverage. This enables comparison of how different outlets present the same event and reveals hidden biases in long-term reporting.

Discourse Analysis

It enables structured analysis of **public discourse evolution**, showing how narratives change in response to real-world developments. By tracking these shifts, researchers can understand how societal concerns, priorities, and interpretations develop over time, making it easier to study large-scale communication patterns.

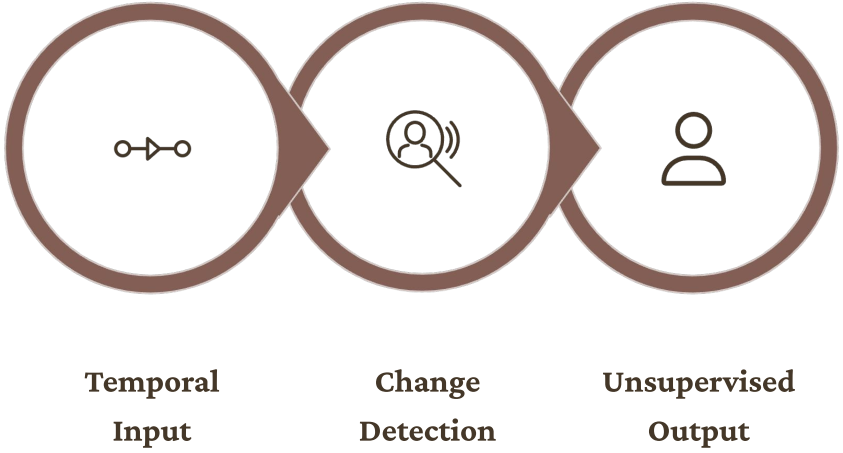
Misinformation Detection

Narrative shift detection can identify **abnormal or sudden changes in framing**, which may indicate misinformation or manipulation. By detecting inconsistencies in how events are described, the system helps flag misleading narratives and supports efforts to monitor information reliability at scale.

Policy Analysis

By linking narrative changes to real-world events, narrative shift detection provides **evidence-based insights for policy and decision-making**. It helps policymakers understand how issues are perceived over time and how media narratives influence public response, enabling more informed and timely decisions.

Problem Statement



Input

A chronologically ordered sequence of news articles spanning multiple time windows.

Output

Detection of **when** a shift occurred and the sentences responsible for the shift.

Constraint

Fully **unsupervised** — no labeled data, predefined shift annotations, or domain-specific rules.

Objectives of the Work

01

Topic-Agnostic Framework

Develop a unified system capable of detecting narrative shifts across **diverse domains such as war, climate, health, economics, and technology** without requiring topic-specific retraining. The goal is to ensure generalization across heterogeneous data sources and enable scalable analysis of narratives in any domain.

02

Unsupervised Learning

Design a fully **self-supervised framework using Temporal Contrastive Learning (TCL)** that eliminates dependence on manually labeled data. The model learns narrative patterns directly from temporal relationships between text segments, making it suitable for large-scale, real-world datasets.

03

Sentence-Level Interpretability

Provide **fine-grained explanations of narrative shifts** by identifying specific sentences that drive changes in meaning. This ensures transparency and allows users to understand not just *when* a shift occurs, but also *what linguistic elements caused it*.

04

Scalability

Build a computationally efficient pipeline capable of processing **millions of news articles and millions of sentences**. The system should handle large-scale temporal data while maintaining performance, enabling practical deployment for continuous media monitoring.

Data Sources

1.17M

News Articles

Collected from multiple sources spanning over a decade

15

Years Covered(2010-2025)

Strong temporal coverage enabling longitudinal analysis

Collection Strategy

Articles were sourced from **Kaggle datasets, public APIs, and web-based collections** to maximize topical breadth and temporal range.

> Multi-source aggregation is intentional: drawing from diverse outlets reduces **source-level bias** and ensures that detected narrative shifts reflect genuine discourse change rather than outlet-specific editorial patterns.

Data Cleaning & Preprocessing

Preprocessing Pipeline

Raw data contains substantial **noise** — including duplicate articles, missing publication dates, and non-English text — that must be removed before modeling.

The pipeline filters approximately **70% of raw content**, retaining only articles that meet quality thresholds for language, date availability, and content length.

> The resulting **~355,000 high-quality articles** form a reliable and temporally well-structured corpus for downstream analysis.

Sentence-Level Processing

Article Segmentation

Each article is segmented into individual sentences, enabling **fine-grained analysis** at the sub-document level rather than treating each article as a monolithic unit.

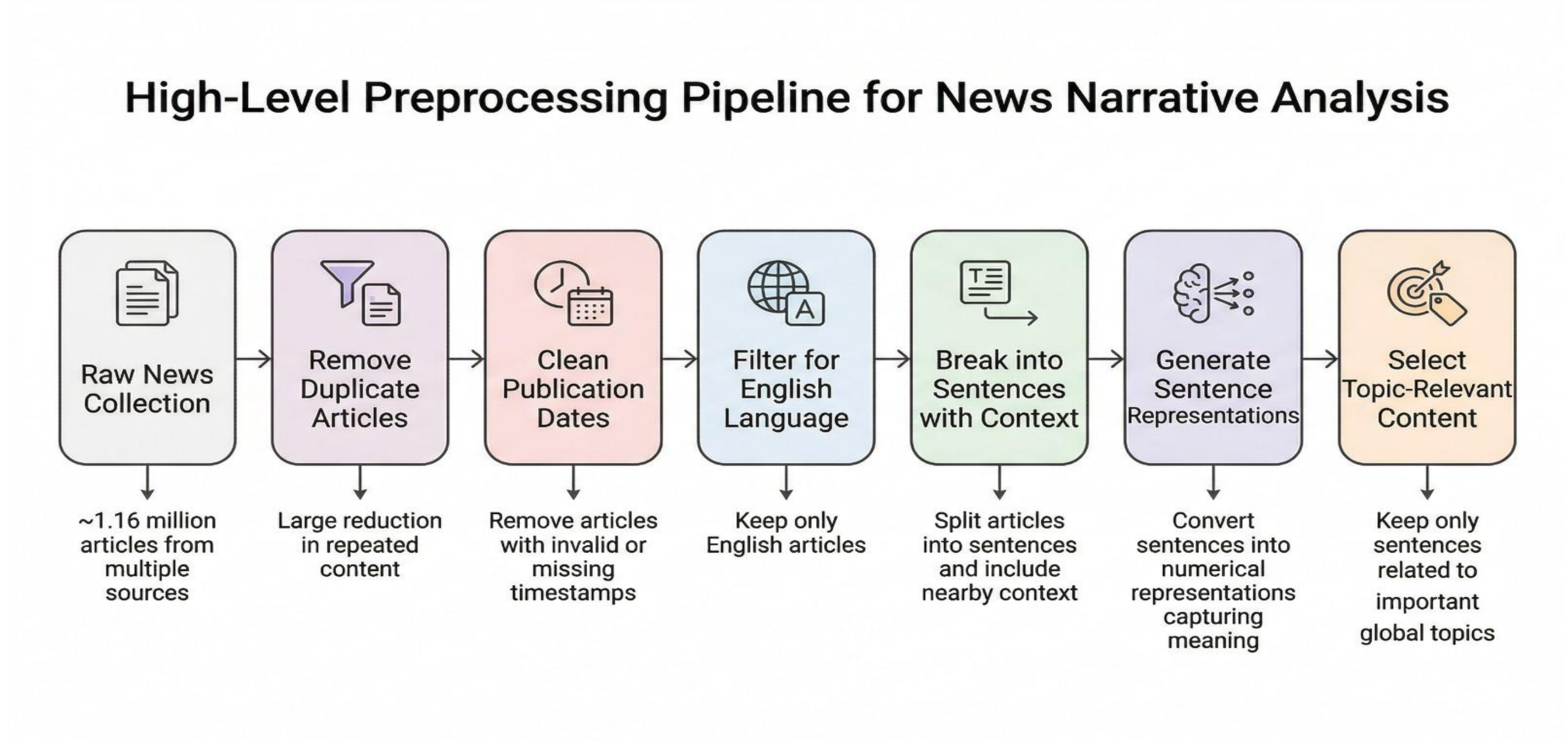
SBERT Embeddings

Context-aware sentence embeddings are generated using Sentence-BERT (SBERT), preserving semantic meaning and contextual nuance for downstream contrastive learning.

Filtering & Retention

From approximately **3 million processed sentences**, around 1.3 million topic-relevant sentences are retained — ensuring the model trains only on high-signal content.

Data Cleaning & Preprocessing



SBERT Baseline Approach

Leveraging dense semantic embeddings to detect shifts in narrative meaning across time.

Semantic Embeddings

Sentence-BERT (SBERT) generates **context-aware, dense vector representations (768-D)** for each sentence, capturing semantic meaning rather than surface-level keywords. In this project, contextual windows (e.g., w1, w3, w5) are used to incorporate surrounding sentences, improving representation of narrative flow.

Similarity Measurement

Narrative similarity is computed using **cosine similarity between sentence or window embeddings**, enabling comparison across time. Changes in similarity reflect **semantic drift**, indicating how the meaning and framing of narratives evolve in high-dimensional embedding space.

Shift Detection

Narrative shifts are detected by measuring **embedding drift between consecutive time windows**. Significant divergence (above calibrated thresholds) indicates a potential shift in narrative framing or interpretation.

Limitations of SBERT

Static by Design

- Produces fixed, pre-trained embeddings
- Does not adapt to evolving narrative context
- Cannot capture dynamic meaning changes over time

No Temporal Modeling

- Treats each sentence independently
- Ignores temporal order and sequence information
- Fails to model narrative progression and continuity

Limited Shift Detection Capability

- Relies only on embedding drift
- Cannot distinguish true narrative shifts from noise
- Weak separation between meaningful change and variation

Surface Sensitivity

- Sensitive to minor wording or stylistic differences
- Small changes may trigger false positives
- Misinterprets linguistic variation as narrative shifts

Lacks Interpretability

- Provides only similarity scores
- Cannot explain why a shift occurred
- Does not identify sentences responsible for change

Fixed Window Limitation

- Uses rigid temporal windows (e.g., 5-day bins)
- May miss real shift boundaries
- Cannot adapt to different narrative speeds

K-Means Clustering Approach

Semantic Grouping

Sentence embeddings are grouped into **K clusters (K=5)**, where each cluster represents a distinct semantic pattern or narrative theme derived from SBERT embeddings.

Cluster as Structure

Each cluster acts as a proxy for an underlying narrative structure, and every time window is represented as a distribution over these clusters to capture dominant themes.

Distribution Shifts

Narrative shifts are identified by observing changes in cluster membership over time, where movement of data points across clusters reflects evolving narrative focus.

Jensen-Shannon Divergence

Divergence between cluster distributions is quantified using Jensen-Shannon divergence, where higher divergence values indicate stronger narrative shifts across time windows.

Limitations of K-Means Clustering Approach

Fixed Cluster Count

- Requires predefined number of clusters (K=5)
- May not reflect real narrative complexity
- Not adaptable across topics and time

Ignores Temporal Order

- Operates on unordered data
- Does not consider sequence or time dependency
- Fails to capture narrative evolution

No Sentence Attribution

- Cannot identify shift-causing sentences
- Lacks fine-grained interpretability
- Provides only cluster-level insights

Limited Narrative Insight

- Captures only distribution-level changes
- Provides static thematic grouping
- Does not model dynamic narrative flow

Weak Temporal Sensitivity

- Cannot distinguish gradual vs abrupt shifts
- Not time-aware in clustering process
- Leads to inaccurate shift detection

Dependence on Initialization

- Sensitive to initial centroid selection
- Results may vary across runs
- Can produce unstable clustering

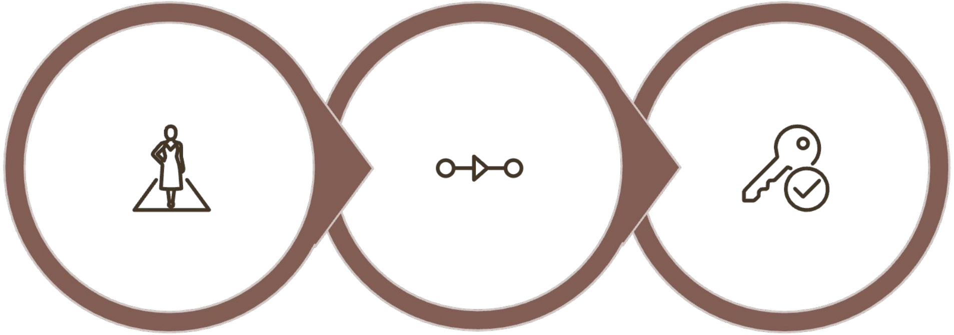
CORE INSIGHT

Narrative Shift Is a Temporal Problem

Static models fundamentally fail at narrative shift detection because they treat language as a frozen snapshot — ignoring the fact that meaning is constructed and evolves across time.

To detect shifts accurately, a model must **explicitly represent temporal relationships** between segments, understanding not just what is said, but when and in relation to what came before.

This critical gap motivates the adoption of **Temporal Contrastive Learning** as a more principled, time-aware framework.



Static Models Fail

Temporal Relationships Matter

TCL Solves It

Temporal Contrastive Learning (TCL)

Time-Ordered Learning

TCL is a **representation learning framework** specifically designed to process and understand sequences of data ordered across time.

Cross-Temporal Comparison

The model learns by **contrasting samples** from different time steps, building awareness of how meaning shifts across temporal boundaries.

Continuity and Change

Adjacent segments are treated as similar while distant segments are pushed apart — teaching the model to detect both continuity and meaningful divergence.

Contrastive Learning Principle

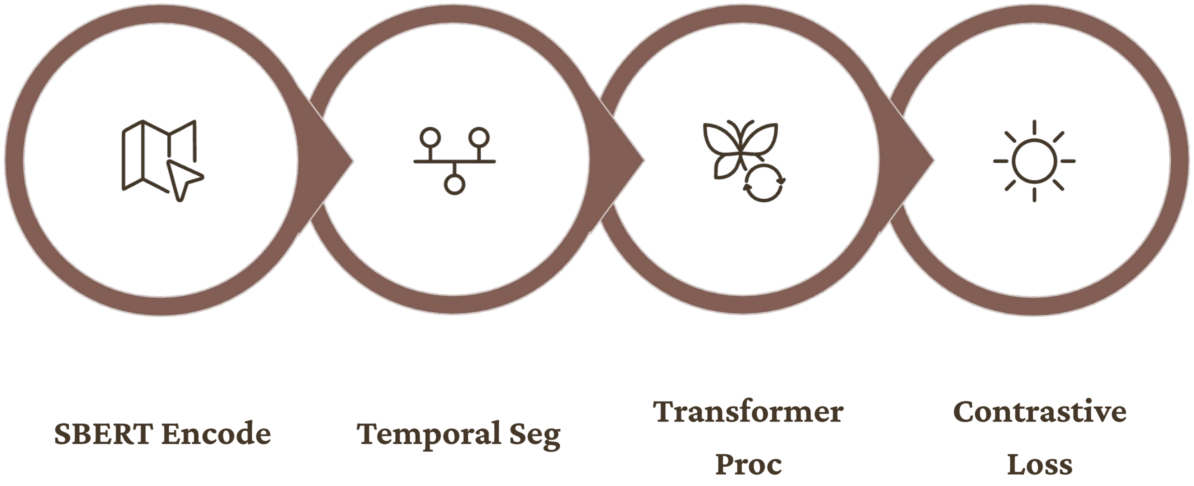
How the Embedding Space Is Shaped?

The contrastive learning objective directly sculpts the geometry of the embedding space to reflect narrative structure.

- **Pull similar samples closer** — embeddings from semantically continuous segments are drawn together in representation space.
- **Push dissimilar samples apart** — embeddings from distant or divergent segments are separated, creating clear cluster boundaries.
- **Shifts surface as distance** — narrative transitions manifest as large geometric gaps between consecutive embeddings, making them readily detectable.

TCL Pipeline

A four-stage architecture that transforms raw sentences into temporally structured, shift-aware representations.



Stage 1 — Encode
Raw sentences are converted into **dense embeddings** using SBERT as the foundational encoder.

Stage 2 — Segment
Embeddings are organized into **ordered temporal windows** that preserve sequential structure.

Stage 3 — Transform
A **transformer encoder** processes each segment, integrating contextual information across the sequence.

Stage 4 — Learn
Contrastive loss is applied to enforce temporal proximity and separation across the representation space.

Temporal Loss Concept

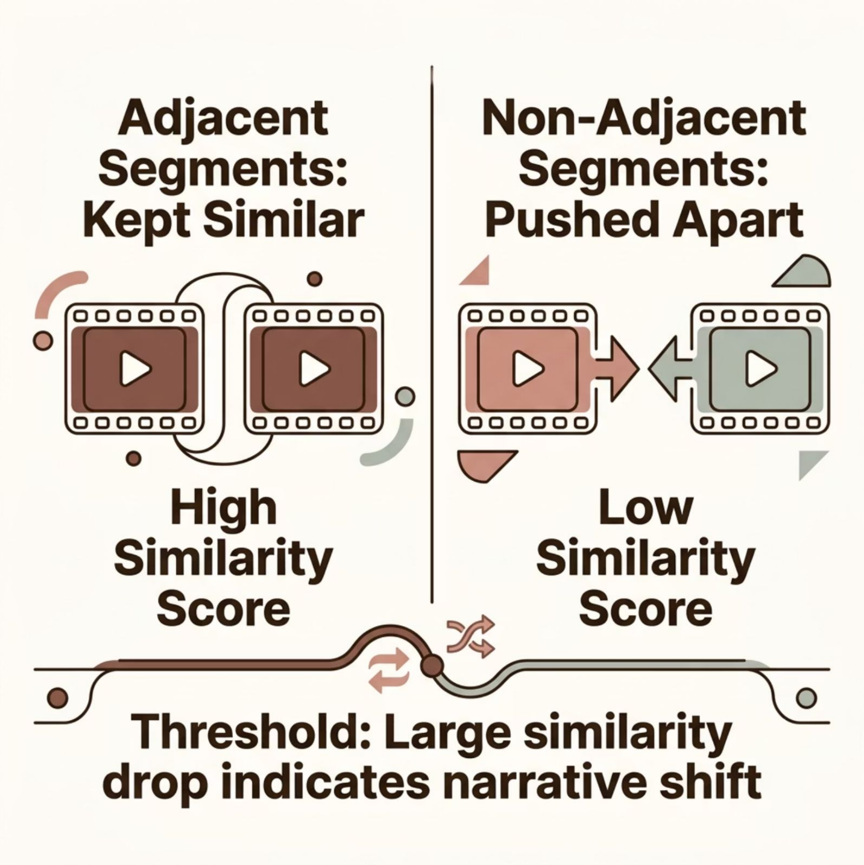
Shaping the Learning Signal

Temporal contrastive loss is designed to explicitly model narrative evolution by learning relationships between temporally ordered segments. It ensures that representations capture both continuity and meaningful change across time.

Adjacent segments are penalized for drifting apart, ensuring the model learns smooth, meaningful continuity.

Non-adjacent segments are penalized for being too similar, preserving discriminative structure across the sequence.

A **sudden drop in similarity** between consecutive segments serves as the primary signal for narrative shift detection.



Advantages of Temporal Contrastive Learning

Temporal Awareness

TCL explicitly models temporal relationships by learning from sequential segments, enabling the system to capture both gradual evolution and abrupt narrative shifts rather than treating text as independent data points.

Unsupervised by Design

The framework operates without labeled data by leveraging contrastive learning objectives, making it scalable and adaptable to large, real-world news corpora without manual annotation.

Richer Structure

TCL learns a well-structured embedding space where temporally related segments are grouped together while distinct narrative states are separated, resulting in more meaningful representations than static methods like SBERT or K-Means.

Accuracy & Interpretability

By combining temporal modeling with contrastive objectives, TCL improves shift detection accuracy and provides interpretable outputs by identifying when shifts occur and which sentences contribute to those changes.

METHODOLOGY

Need for Multiple Approaches

Initial models fell short of capturing the full complexity of narrative behavior — prompting an iterative exploration of progressively stronger approaches.

1. Insufficient Baseline

Initial baseline methods such as SBERT drift and K-Means clustering were unable to capture the **temporal dynamics and semantic complexity** of narrative evolution, often missing subtle shifts or producing noisy detections.

2. Broad Exploration

Multiple approaches were explored, including **window-based modeling, grouping strategies, and contrastive learning**, to systematically address limitations and improve robustness across diverse topics.

3. Iterative Refinement

Each successive approach introduced targeted improvements such as **noise reduction, adaptive grouping, and semantic segmentation**, enabling better alignment with real narrative behavior over time.

4. Robust Final Model

The final TCL-based framework integrates **temporal modeling, topic separation, and adaptive segmentation**, resulting in a scalable and generalizable solution for accurate narrative shift detection.

APPROACH 1

Fixed Windows

Temporal Grouping

- Sentences grouped into fixed-size windows (e.g., overlapping/non-overlapping)
- Creates discrete temporal segments for modeling narrative flow
- Overlapping windows improve continuity but may introduce redundancy

Transformer Encoding

- Each window encoded using transformer-based model
- Produces dense, context-aware representations (128-D output)
- Incorporates time-gap and topic features for better structure

Contrastive Learning (TCL)

- Adjacent windows treated as positive pairs (similar)
- Non-adjacent windows treated as negative pairs (dissimilar)
- Uses InfoNCE / NT-Xent loss to learn temporal relationships

Shift Detection Mechanism

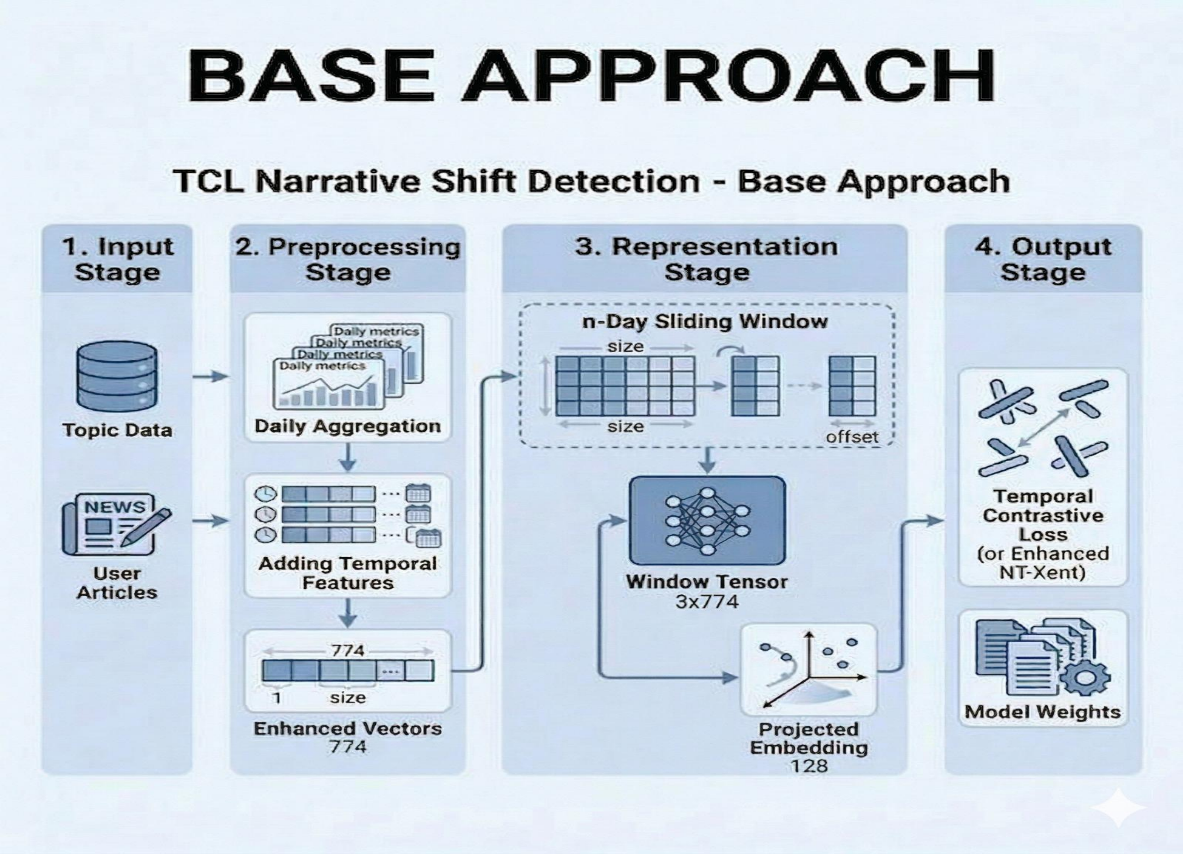
- Narrative shifts identified using similarity drop between windows
- High loss or low similarity indicates potential shift
- Captures continuity vs discontinuity in narrative evolution

Baseline Characteristics

- Simple and structured starting point for TCL framework
- Demonstrates importance of temporal modeling over static methods
- Serves as reference for evaluating improved approaches

APPROACH 1

Fixed Windows



Limitations of Approach 1

No Adaptive Speed

- Fixed window size cannot adapt to different narrative evolution rates
- Fast-changing topics require smaller windows, slow topics need larger ones
- Leads to misalignment with real narrative boundaries

Noise Sensitivity

- Highly sensitive to small variations in sentence content
- Minor semantic changes can distort window representations
- Results in unstable learning and inconsistent shift detection

Lack of Semantic Awareness

- Grouping based only on time, not meaning
- Ignores underlying semantic changes within windows
- Limits ability to capture fine-grained narrative shifts

Poor Generalization

- Limited ability to handle diverse topics without strong separation
- Simple topic encoding (one-hot) is not expressive enough
- Model struggles across domains with different narrative patterns

Fixed Segmentation Issue

- Windows may mix multiple narrative states within a single segment
- Cannot detect true semantic boundaries
- Reduces accuracy of shift localization

False Shift Detection

- Lack of shared context in some window setups causes abrupt differences
- Model may interpret noise or stylistic variation as real shifts
- Leads to high false positive rates

APPROACH 2

Grouping Strategy

Stable Units

- Sentences grouped into stable semantic units before modeling
- Reduces impact of sentence-level noise and variation
- Creates more meaningful temporal representations

Mean Pooling

- Aggregates sentence embeddings using mean pooling
- Produces compact group-level representations
- Smooths minor variations across sentences

Noise Reduction

- Grouping minimizes effect of noisy or inconsistent sentences
- Handles missing or sparse data across time
- Improves stability of temporal signals

Smoother Representations

- Generates smoother transitions between time segments
- Better captures gradual narrative evolution
- Reduces false shift detection caused by abrupt changes

Improved Temporal Consistency

- Groups preserve local temporal continuity
- Prevents fragmentation of narrative flow
- Enhances learning of sequential relationships

Handling Data Sparsity

- Robust to irregular time gaps in data
- Avoids broken windows due to missing days
- Maintains consistent temporal grouping

APPROACH 2

Improvements in Approach 2

Noise Dampening

- Grouping reduces influence of outlier or noisy sentences
- Mean pooling smooths small variations in embeddings
- Leads to more stable and reliable representations

Handles Irregular Data

- Robust to missing or unevenly spaced time data
- Prevents broken temporal windows due to gaps
- Maintains continuity in real-world datasets

Temporal Consistency

- Group-level embeddings preserve local temporal continuity
- Reduces abrupt fluctuations between consecutive segments
- Better reflects gradual narrative evolution

Reliable Shift Detection

- Reduces false positives caused by noise or sparsity
- Improves stability of similarity-based shift signals
- Enables more accurate identification of narrative transitions

Improved Representation Quality

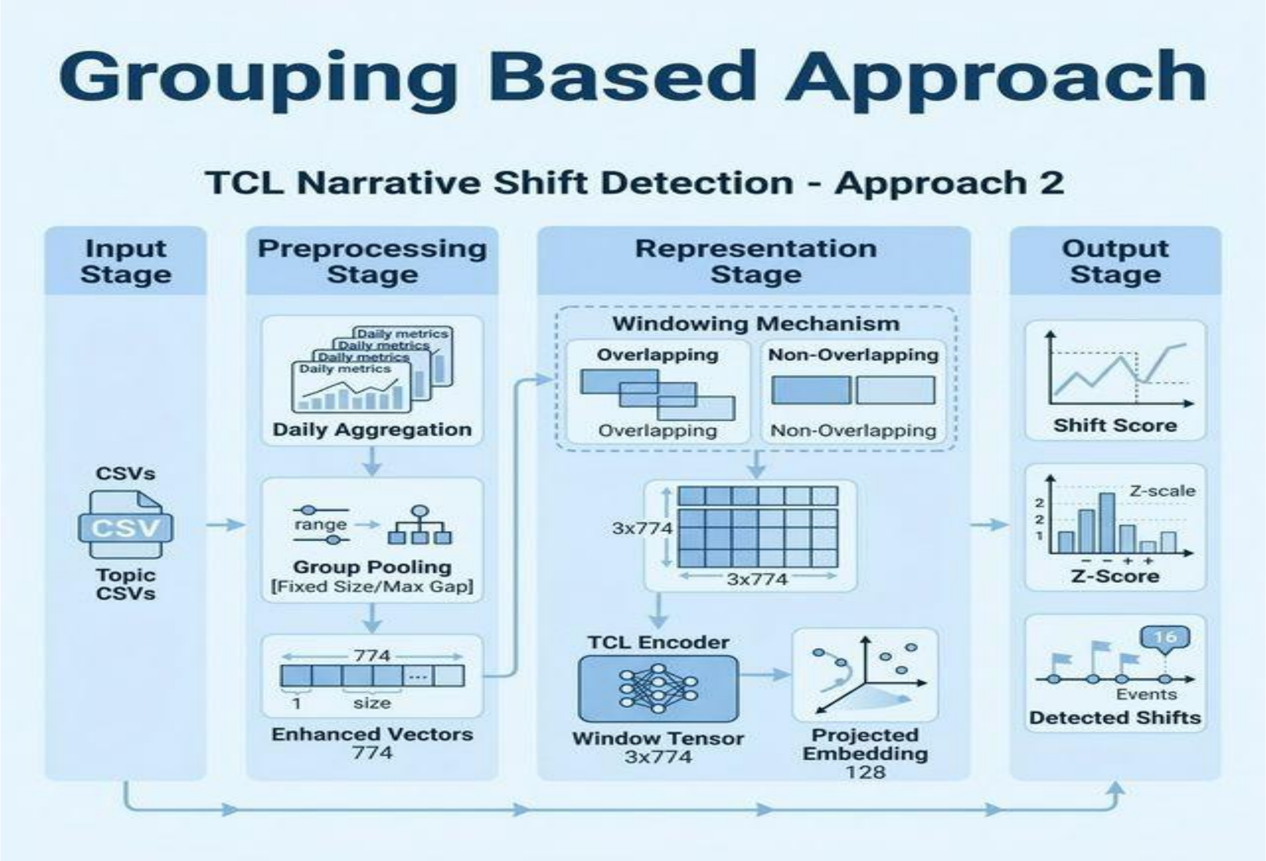
- Aggregated embeddings capture richer contextual information
- Less sensitive to individual sentence anomalies
- Enhances overall semantic coherence

Better Learning Stability

- Contrastive learning becomes more stable with grouped inputs
- Loss decreases more consistently during training
- Reduces variance across training runs

APPROACH 2

Improvements in Approach 2



APPROACH 2

Limitations in Approach 2

Rule-Based Grouping

- Relies on predefined heuristics (fixed-size or max-gap grouping)
- Not learned from data → lacks adaptability
- Requires manual tuning of thresholds

No Semantic Awareness

- Grouping based only on time proximity
- Does not consider actual semantic change
- May hide true narrative boundaries within groups

Topic Mixing

- Different narratives or topics may be merged in same group
- Mean pooling blends distinct semantic signals
- Reduces clarity of shift detection

Static Logic

- Same grouping rules applied across all topics
- Cannot adapt to different narrative speeds
- Limits generalization across domains

Heuristic Sensitivity

- Performance depends on chosen grouping parameters
- Poor threshold selection degrades results
- Requires experimentation for optimal setup

Loss of Fine-Grained Signals

- Aggregation removes sentence-level variations
- Subtle shifts may get smoothed out
- Reduces sensitivity to early-stage narrative changes

APPROACH 3

Dynamic Windows

Variable Topic Speed

- Different narratives evolve at different temporal rates
- Fixed windows fail to capture this variability
- Requires flexible segmentation aligned with change intensity

Adaptive Window Sizing

- Window size determined dynamically using data-driven signals
- Adjusts based on local semantic change or similarity drop
- Enables context-aware segmentation

Speed-Matched Granularity

- Fast-changing narratives → smaller windows for fine detail
- Slow-changing narratives → larger windows for broader context
- Balances sensitivity and stability in representation

Real-World Alignment

- Better matches actual narrative boundaries in data
- Avoids artificial segmentation caused by fixed windows
- Improves consistency with real temporal structure

Improved Shift Localization

- Detects shifts closer to their true occurrence point
- Reduces delay in identifying narrative transitions
- Enhances temporal precision

Reduced Noise Impact

- Adaptive grouping prevents unnecessary fragmentation
- Avoids splitting stable narratives into multiple segments
- Improves robustness of similarity signals

DECISION POINT

Why Approach 3 Was Not Used?

Key Takeaway

Despite its theoretical appeal, Approach 3 was ultimately set aside in favor of a solution that balances performance with scalability and implementation simplicity.

Computational Cost

- Requires repeated similarity calculations for adaptive window tuning
- Involves iterative refinement of window boundaries
- Significantly increases processing time and resource usage

Scalability Barrier

- Not suitable for large-scale datasets (millions of articles)
- Per-document dynamic segmentation is computationally expensive
- Difficult to deploy in real-time or production settings

Sensitivity to Parameters

- Performance depends on threshold selection for similarity changes
- Requires careful tuning for different datasets
- Reduces robustness across domains

Pipeline Complexity

- Introduces additional steps for dynamic window adjustment
- Makes training and inference pipelines more complex
- Increases risk of implementation and debugging challenges

Practical Limitations

- Hard to integrate into scalable pipelines
- Not ideal for continuous monitoring systems
- Motivated need for simpler, more deployable approach

Inconsistent Window Sizes

- Produces variable-length segments across documents
- Makes batching and model training less efficient
- Leads to irregular representation structures

APPROACH 4

Semantic Segmentation

Semantic-Driven Boundaries

- Segmentation based on actual semantic change, not fixed heuristics
- Uses embedding similarity to detect meaningful transitions
- Aligns segments with true narrative structure

Fully Data-Driven

- No reliance on manual rules or predefined grouping heuristics
- Learns segmentation directly from data patterns
- Generalizes well across different domains

Adaptive Segments

- Segment sizes vary dynamically based on semantic continuity
- Captures both gradual evolution and abrupt changes
- Produces more meaningful and flexible representations

Change-Point Detection

- Identifies precise points where narrative meaning shifts
- Uses similarity drop or contrastive signals as indicators
- Enables accurate localization of narrative transitions

Improved Shift Detection

- Detects shifts closer to true semantic boundaries
- Reduces false positives from arbitrary segmentation
- Enhances overall detection accuracy

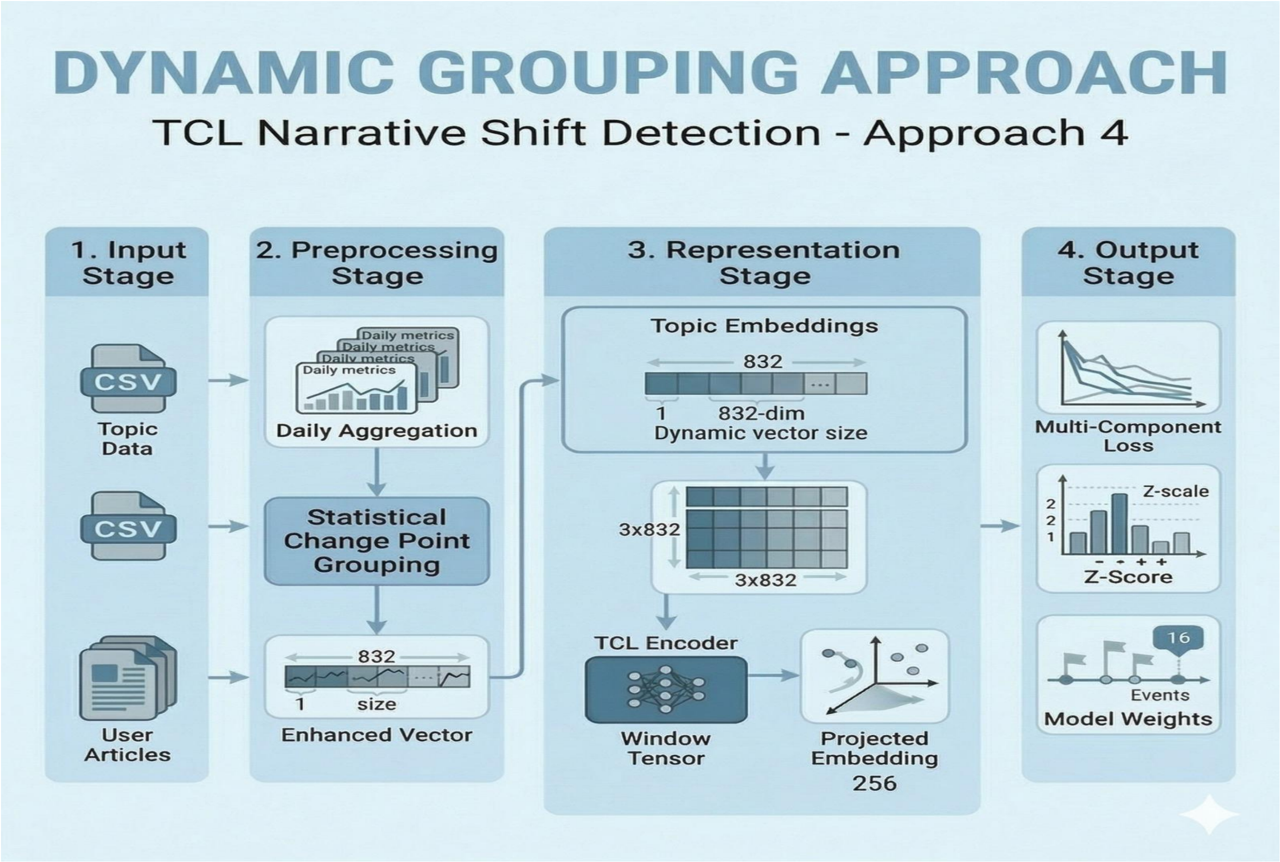
Better Interpretability

- Provides clear boundaries linked to semantic changes
- Easier to trace which segments caused the shift
- Supports sentence-level explanation of results

Robust to Noise

- Ignores minor fluctuations within stable segments
- Focuses on significant semantic transitions

Semantic Segmentation



Multi-Loss Learning

Temporal Loss

- Encourages similarity between adjacent segments
- Preserves sequential continuity of narratives
- Penalizes sudden, unjustified embedding drift

Topic Separation Loss

- Pushes apart embeddings from different topics
- Prevents mixing of unrelated narrative threads
- Improves clustering of distinct semantic groups

Hard Negative Loss

- Focuses on difficult negative samples (semantically close but different)
- Sharpens decision boundaries in embedding space
- Reduces ambiguity between similar narratives

Balanced Optimization

- Reduces ambiguity between similar narratives
- Ensures trade-off between continuity and discrimination
- Stabilizes learning across diverse data distributions

Improved Representation

- Produces structured and well-separated embedding space
- Captures both temporal flow and semantic distinction
- Enhances downstream shift detection performance

Combined Effect

- Enhances downstream shift detection performance
- Generates robust and discriminative representations
- Enables accurate and interpretable narrative shift detection

Why Approach 4 performs the best ?

Adaptive Segmentation

- Segmentation based on real semantic change, not fixed rules
- Dynamically adjusts to narrative evolution speed
- Produces meaningful and context-aware segments

True Narrative Boundaries

- Detects boundaries where actual meaning shifts occur
- Avoids artificial cuts from fixed or heuristic methods
- Aligns segmentation with real-world narrative structure

Top Evaluation Metrics

- Achieves highest performance across evaluation benchmarks
- Outperforms SBERT, K-Means, and earlier TCL approaches
- Demonstrates consistent improvement in shift detection accuracy

Continuity & Separation Balance

- Maintains smooth narrative continuity within segments
- Ensures clear separation between different narrative states
- Solves trade-off between over-segmentation and under-segmentation

Better Shift Localization

- Identifies shifts closer to their true occurrence point
- Reduces delay in detection compared to earlier methods
- Improves temporal precision of results

Robust Across Domains

- Generalizes well across topics like geopolitics, health, and technology
- Handles diverse narrative structures effectively
- Works without domain-specific tuning

APPROACH 5

Entity Awareness

Named Entity Extraction (NER)

- Uses NER to extract key entities (people, locations, organizations)
- Entities act as semantic anchors in narrative analysis
- Captures important actors driving the narrative

Entity-Aware Representation

- Combines sentence embeddings with entity information
- Enhances semantic representation with contextual signals
- Improves distinction between similar textual content

Entity Overlap Analysis

- Measures overlap of entities across consecutive segments
- High overlap → narrative continuity
- Low overlap → potential narrative shift

Fine-Grained Shift Detection

- Detects shifts within same topic using entity-level changes
- Identifies subtle transitions missed by pure semantic models
- Captures evolving focus of narratives

Improved Interpretability

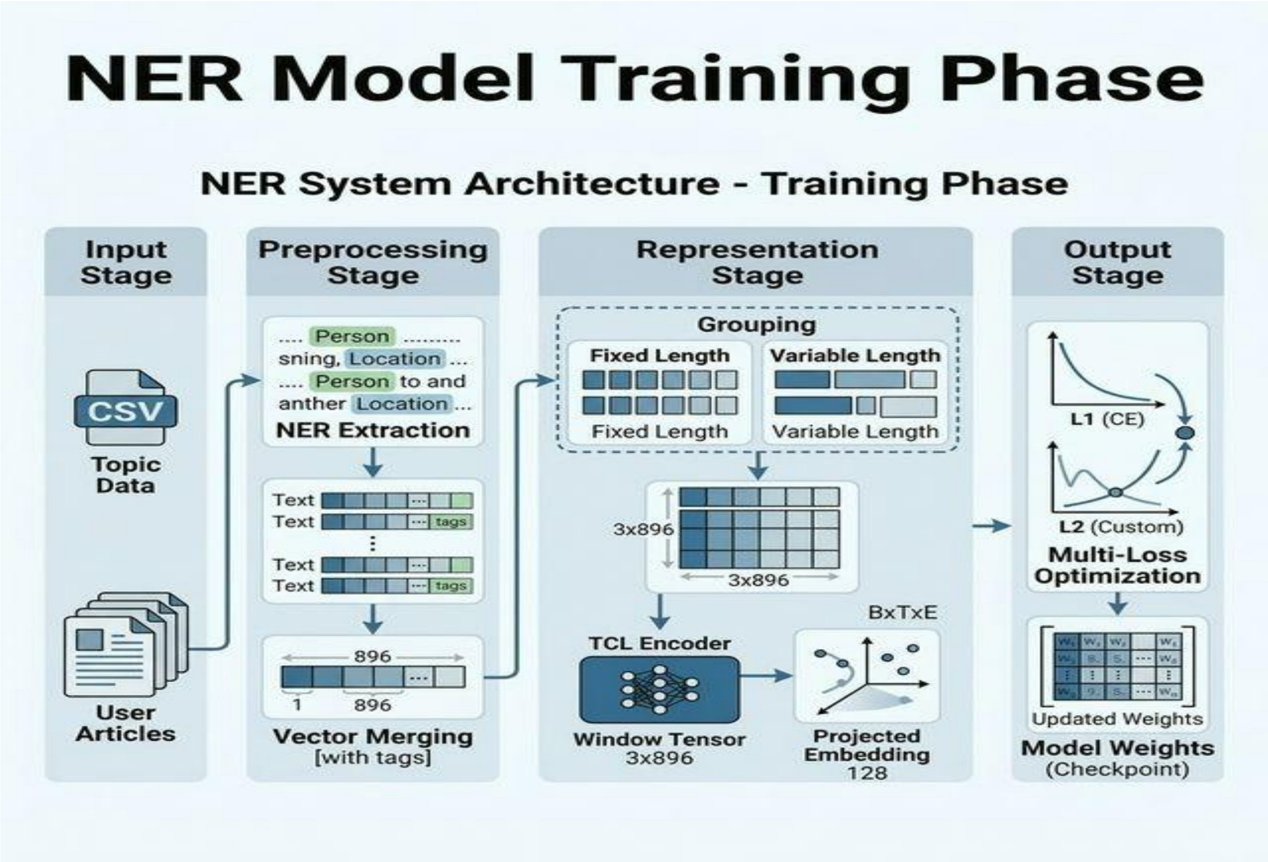
- Enables tracing which entities contributed to a shift
- Provides explainable reasoning for model decisions
- Useful for analysis in media and policy contexts

Topic Disambiguation

- Helps separate narratives with similar language but different actors
- Reduces confusion between overlapping topics
- Strengthens semantic clarity

APPROACH 5

Entity Awareness



Limitations in Approach 5

Multi-Loss Balancing

- Multiple loss functions introduce competing objectives
- Difficult to tune weights between temporal, topic, and hard-negative losses
- Improper balance can degrade overall performance

Training Instability

- Conflicting gradients from different losses affect convergence
- Leads to unstable training and oscillating loss behavior
- Requires careful optimization strategies

Inconsistent Gains

- Performance improvements vary across datasets and domains
- Benefits not uniform for all narrative types
- Reduces reliability in real-world deployment

Further Optimization Needed

- Requires additional tuning of architecture and hyperparameters
- Needs better integration of entity and semantic signals
- Not yet fully optimized compared to Approach 4

Increased Computational Cost

- Additional NER processing adds overhead
- Multi-loss training increases computation time
- Less efficient compared to simpler models

Dependency on NER Quality

- Performance depends on accuracy of entity extraction
- Errors in NER can propagate to final predictions
- May miss implicit or ambiguous entities

Quantitative Results

Final Insight

Combining semantic awareness with temporal modeling leads to best results

Approach 4 successfully balances continuity and separation

Establishes a strong foundation for practical narrative shift detection

Overall Performance Comparison

- Approach 4 achieves the highest score across all evaluated methods
- Consistently outperforms SBERT, K-Means, and intermediate approaches
- Demonstrates clear improvement over both static and heuristic baselines

Intra-Topic Similarity

- Highest intra-topic similarity indicates strong semantic coherence within segments
- Embeddings within each segment remain tightly grouped
- Reflects accurate preservation of narrative continuity

Temporal Consistency

- Maintains smooth transitions between adjacent segments
- Avoids abrupt or noisy fluctuations seen in earlier approaches
- Captures gradual evolution of narrative structure over time

Separation of Distinct Topics

- Achieves better separation between unrelated narrative segments
- Reduces overlap between different topics compared to baselines
- Improves clarity in identifying true narrative boundaries

Impact of Adaptive Segmentation

- Adaptive windowing contributes most to performance gains
- Enables alignment with real narrative structure instead of fixed boundaries
- Key factor behind improved detection accuracy

Comparison with Baselines

- SBERT baseline struggles with temporal awareness despite strong semantics
- K-Means captures structure but lacks sequential understanding
- Intermediate approaches improve partially but fail to balance all aspects

Key Observations

1. **Temporal Modeling Is Essential**

Capturing how narratives evolve over time is fundamental to accurate shift detection, as meaningful changes often emerge through gradual transitions rather than isolated sentence differences. Models that ignore temporal order treat text as unordered data, which leads to loss of contextual continuity and significantly weaker performance in identifying true narrative shifts.

2. **Adaptive Segmentation Drives Gains**

Using semantically grounded, adaptive segment boundaries allows the model to align more closely with the natural structure of the narrative, rather than relying on arbitrary fixed windows. This results in more precise detection of meaningful transitions, as segments expand or contract based on the pace of topic evolution, leading to consistently higher performance across evaluation metrics.

3. **Multi-Loss Learning Enhances Representations**

Using semantically grounded, adaptive segment boundaries allows the model to align more closely with the natural structure of the narrative, rather than relying on arbitrary fixed windows. This results in more precise detection of meaningful transitions, as segments expand or contract based on the pace of topic evolution, leading to consistently higher performance across evaluation metrics.

4. **Static Models Are Insufficient**

Approaches that rely on static embeddings or fixed grouping strategies are unable to capture the dynamic and evolving nature of real-world narratives. Without mechanisms to model temporal dependencies or semantic transitions, such methods often produce noisy or misleading signals, limiting their effectiveness in practical narrative shift detection tasks.

Case Study — Real-World Validation

Scenario: Conflict Narrative Tracking

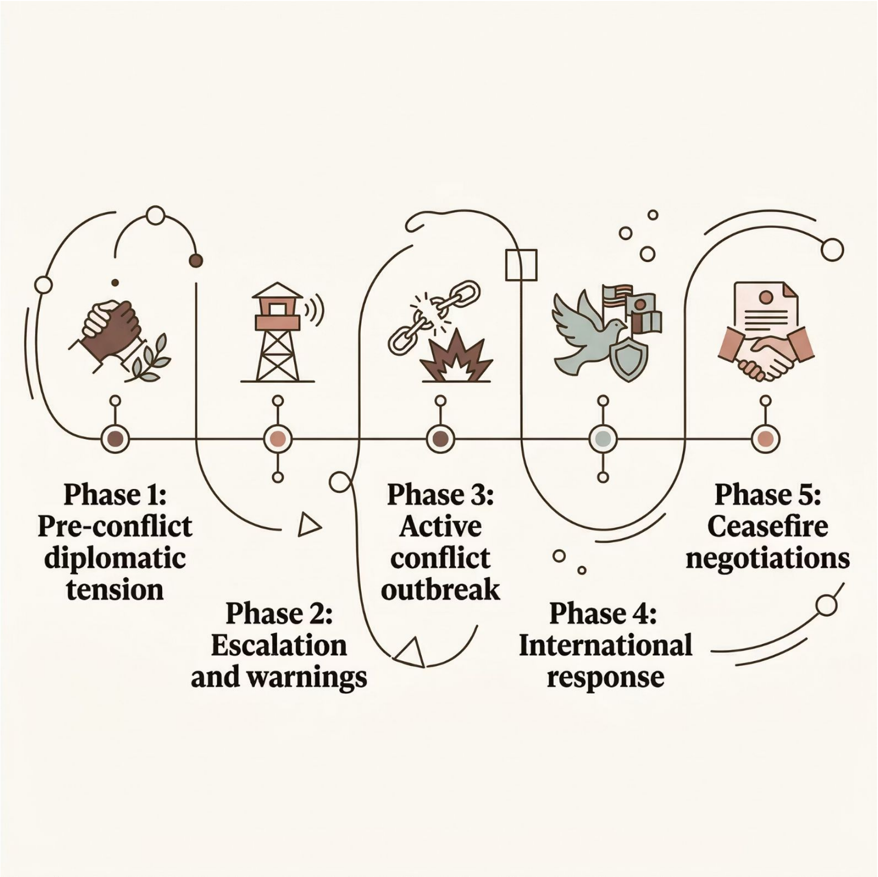
Objective: Evaluate the model on real-world geopolitical news data to test its ability to detect meaningful narrative shifts.

Key Observations

- The model successfully identified the transition from *pre-conflict tension* to *active armed engagement* as a clear narrative boundary.
- It captured both **abrupt shifts** (e.g., sudden outbreak of conflict) and **gradual changes** (e.g., slow diplomatic deterioration).
- Detected segment boundaries closely matched **human-annotated ground truth**, ensuring reliability.

Key Outcome

- The results demonstrate strong **real-world applicability** beyond controlled benchmarks.
- Confirms that the model can effectively track evolving narratives in live, complex data environments.



Error Analysis

Understanding where and why the model fails is critical for guiding future improvements.

1

False Positives

Stylistic or tonal changes — such as shifts in journalistic framing or authorial voice — are sometimes misclassified as genuine narrative shifts, inflating detection counts.

2

False Negatives

When narrative transitions are **gradual and subtle**, the model may fail to cross the detection threshold, causing genuine shifts to go undetected.

3

Threshold Sensitivity

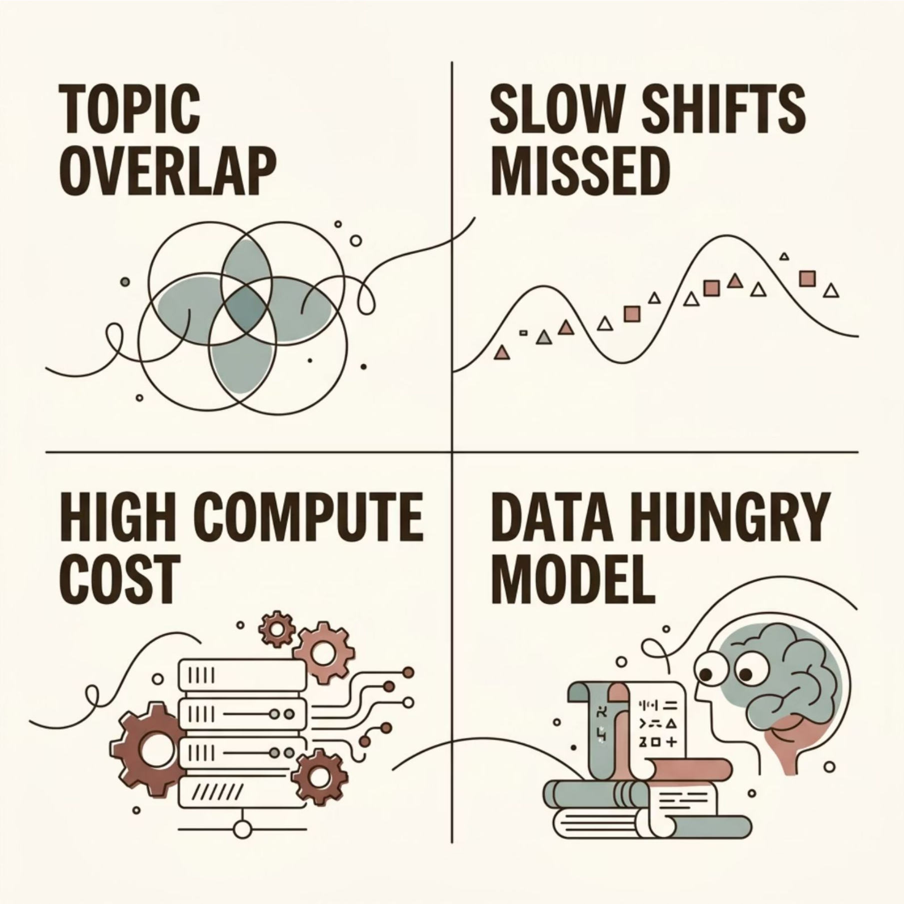
The choice of **detection threshold** significantly affects precision-recall trade-offs; no single value generalizes well across all topic types.

4

Robustness Gaps

Further work is needed to make the model **resilient** to noisy, informal, or low-resource text conditions commonly encountered in real deployments.

Current Limitations



What Remains Unsolved

- Topic separation is not fully robust in semantically adjacent content
- Gradual shifts remain a persistent challenge across datasets
- Computational efficiency needs improvement for real-world deployment
- Dependence on large-scale data limits applicability in sparse domains

Topic separation is not yet perfect — boundary ambiguity persists in thematically adjacent segments.
Gradual shifts remain systematically harder to detect than abrupt, well-defined transitions.
Computational cost is substantial, driven by the overhead of contrastive training at scale.
The model depends on **sufficient historical data**, limiting applicability to low-resource or emerging topics.

Future Work & Summary

The path forward is clear: extend, optimize, and scale this framework for broader real-world impact.

01

Improve Entity Modeling & Multi-Loss Optimization

Refining entity-aware components and resolving **multi-loss conflicts** will stabilize training and yield more consistent performance gains.

02

Extend to Multiple Languages

Adapting the framework to **multilingual corpora** is essential for broad applicability in global, cross-lingual news environments.

03

Enable Real-Time Systems

Optimizing inference speed to support **real-time narrative monitoring** will dramatically increase the model's practical utility in live settings.

0
4

Summary

Temporal contrastive learning is a proven, effective approach for modeling narrative evolution — with clear, actionable paths to further improvement.